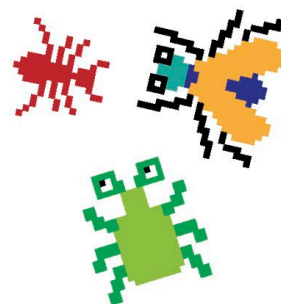


Fixing bugs

If something's wrong with your code, Python will try to help by showing an error message. These messages can seem a bit puzzling at first, but they'll give you clues about why your program isn't working and how to fix it.



Error messages

Both the IDLE editor and the shell window can show error messages if mistakes are detected. An error message tells you what type of error has occurred and where to look in your code.

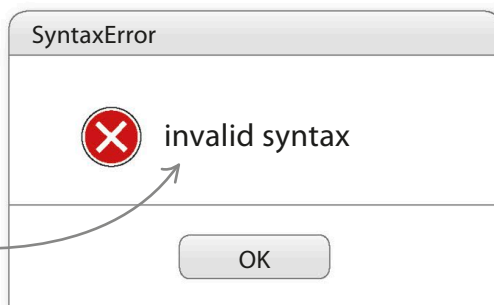
▽ Messages in the shell

Python displays error messages in red text in the shell window. The program stops working when an error message appears. The message tells you which line of code caused the error to happen.

▽ Messages in the IDLE editor

A pop-up box warns you there's an error. Click "OK" to return to your program. There will be a red highlight on or near the error.

This pop-up box tells you there's a syntax error, which means there's a typing mistake.



```
>>>
```

```
Traceback (most recent call last):
```

```
File "Users/Craig/Developments/top-secret-python-book/age.py", line 21, in module>
    print('I am'+ age + 'years old')
```

```
TypeError: Can't convert 'int' object to str implicitly
```

This line tells you it's a type error (see page 50).

The error is on line 21.



EXPERT TIPS

Finding bugs

When an error message appears in the shell, right-click it and choose "Go to file/line" on the drop-down menu. The IDLE editor jumps straight to that line of code so you can start debugging.



I'll find those pesky bugs!

